

Solution for 2025 SAJC Prelim Paper 2

- 1 (a) By Principle of Conservation of Momentum, total momentum of toy car and exhaust gas remains constant since no net external force acts on the system. [1]
- (Since momentum is a vector quantity), exhaust gas gains backward momentum, hence toy car gains forward momentum. [1]
- Rate of change of momentum of car is the force rocket produces on car. [1]
- (b) (i) 1. Any indication of calculating gradient of tangent at 2.0s [1]
 $a = 2.8 \text{ m s}^{-2}$ (accept: range 2.6 to 3.1 m s^{-2}) [1]
2. Net force = ma
Forward force – resistive force = $(0.440)(2.8)$
 $4.6 - \text{resistive force} = (0.440)(2.8)$ [1]
 $\text{resistive force} = 3.368 = 3.4 \text{ N}$ [1]
- (ii) As v increases, gradient of v-t graph, which is the acceleration, decreases. [M1]
(Mass of car is also decreasing.)
Since forward force is constant (at 4.6 N), resistive force must be increasing (so that net force is decreasing). [A1]
- (c) From Fig. 1.2, acceleration due to rocket engine is approximately 8.89 m s^{-2} . This is less than gravitational acceleration of 9.81 m s^{-2} . [M1]
- This means that the rocket engine is unable to propel the car vertically at $t = 0$. [A1]
- However, as the force by rocket engine is constant (at 4.6 N) and car is losing mass (continuously as exhaust gases are produced), eventually acceleration due to rocket force $> 9.81 \text{ m s}^{-2}$ (or force $>$ weight) and the car will be lifted off. [B1]

